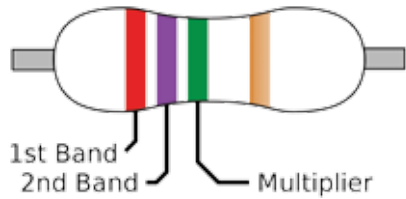


Physical components in an electronic circuit

Resistor

As the name suggests, ‘resistors’, well, resist the flow of current and are used to avoid damage due to high current. It’s denoted by the symbol z and is measured in a unit called ‘Ohm’.



Reading a colour coded resistor

Colour coding is used to denote the value of the resistor. The table here shows colour coding, using which the resistor values are read.

Reading resistor values is important. Let’s see how to do it:

1. Hold the resistor with the SILVER or GOLD coloured band towards the right as shown in the table.
2. Use the table at <http://digit.in/resistorcode> to find the actual value of the resistor.
3. The first two colour bands on the left denote the value of the resistor.
4. The third band denotes the multiplier.
5. The last band denotes the tolerance.

Tolerance is the variation in the value of resistance that can be expected from the resistor you have. Lower tolerance signifies lower variation in resistance and thus proper functioning of the circuit.

In our given figure, the value of resistor is

1st band	2nd band	Multiplier (3rd band)	Tolerance (4th band)
2 (red)	7 (violet)	$\times 10^3$ (green)	$\pm 5\%$

This gives a value of $27 \times 10^3 \pm 5\%$ i.e. $27000 \pm 1350\Omega$.

IC (Integrated Circuit)

When using an ‘integrated circuit’, all connections are made using pin numbers of the IC.

Numbering of IC pins

The most important thing to know is where the 1st pin is located on any IC.

For IC with notch

1. Locate the notch that’s in the shape of a semi-circle.